swig: exact joint placement of V, D and J segments for immune receptor germline assignment

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Abstract

Motivation. Assigning germline V, D and J alleles and segment boundaries to adaptive immune receptor reads is the first step of repertoire analysis, and everything downstream inherits its errors. The D segment is short and trimmed at both ends, somatic hypermutation blurs the evidence that separates alleles, and a junction base that matches two segments can belong to either. Most aligners place the segments one after another and accept a D by a tuned score threshold.

Results. We present swig, a dependency-free Rust aligner that scores every segment hypothesis as a log-likelihood ratio and chooses the alleles and the cut points of V, D and J jointly, by an exact semi-Markov Viterbi recursion over a placement graph. The recursion considers every pair of cuts between neighbouring segments in time linear in their overlaps, and is checked against exhaustive enumeration. The prior cost of each segment is derived as prior odds over the gene-and-position hypotheses it competes with, rather than tuned. A D also pays geometric trim and N/P-length priors, and, at an open read end, the likelihood ratio of the flanking segment the read would have shown. Each read is valued at its own divergence, and forward–backward over the same graph gives posterior probabilities of which the call is a mode. Seeded, banded alignment over gene frames, lossless sibling bounds and an inter-sequence SIMD Gotoh kernel make the search fast. On reads simulated by GenAIRR and scored against its ambiguity-aware truth, swig 0.39.7 has the highest V, D and J accuracy of swig, IgBLAST 1.22 and MiXCR 3.0.13 on every human immunoglobulin set, raising human IGH D accuracy over IgBLAST by 0.7–6.0 percentage points across 0–10 % SHM (0.9–6.3 on reads disjoint from those its D priors were designed on), and assigns about 7,000 reads per second on four threads, 76–88 times the throughput of IgBLAST.

Availability. swig is open source under the MIT licence at `https://github.com/mashu/swig` (crate `engine/`); it builds with Rust $\geq$ 1.89, has no dependencies, and compiles to native code and to WebAssembly.

1 Introduction

High-throughput sequencing of B- and T-cell receptor repertoires (AIRR-seq) turns millions of rearranged receptor transcripts into reads. Each read is a mosaic: a V, a D (heavy chains and TRB/TRD only) and a J gene segment, trimmed by exonucleases at the joints and separated by non-templated N and palindromic P nucleotides, and, in B cells, overlaid with somatic hypermutation (SHM). Germline assignment names the allele of each segment and locates its boundaries, which fixes the junction (CDR3). Clonal grouping, mutation analysis, lineage reconstruction and inference of novel alleles all start from these calls, so their errors propagate.

Three properties make the assignment uncertain rather than merely expensive. First, human IGHD genes are 11–37 nucleotides long and lose several bases at each end; a remnant of 4–8 nucleotides is common, and random sequence matches a few bases of some D gene by chance at many junction positions. Second, under SHM a mismatch may be a mutation, a sequencing error

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or the base that separates two alleles. Third, an N base that happens to match the neighbouring germline can be explained by either segment, so where V ends, where D starts and where J starts are decided together. The task is therefore to choose the most probable explanation among many partial ones, and to say how probable it is.

Widely used tools approach it in different ways. IgBLAST [1] and IMGT/V-QUEST [2] align the read to V germlines and then search for D and J in the remaining sequence. MiXCR [3] uses k-mer seeding and alignment for speed. partis [4] annotates with hidden Markov models whose parameters are inferred from the repertoire, at a much higher computational cost. Benchmarks with an ambiguity-aware simulator [5] show that these aligners make distinct errors, most of them in D. Deciding segments in sequence lets a neighbour's local alignment absorb a short D, and fixed score thresholds for accepting a D trade sensitivity on short remnants against spurious D calls on D-free junctions in ways that are hard to interpret.

swig instead defines one objective, a log-likelihood ratio over the whole read, and maximises it exactly over the segment alleles and every pair of cut points between neighbouring segments. Its contributions are:

1. a *placement graph* over candidate V, D and J alignments whose semi-Markov Viterbi recursion is exact over all grammatical paths and all cut pairs, in time linear in the overlaps between placements (Section 3);

2. segment priors *derived* as prior odds over the hypotheses a segment competes with, D trim and N/P-length priors that re-rank D alleles without changing how often a typical D is called, and an *open-flank* cost for a D at a truncated read end (Section 2.4);

3. calls valued at each read's own divergence in exact integer arithmetic, with forward–backward posteriors over the same graph, so the call is a mode of the reported distribution (Sections 3.3 and 3.4);

4. a search whose every shortcut is labelled with its guarantee (exact, lossless, bounded, defined, approximate or heuristic), including lossless sibling bounds over gene frames (Section 5), and an inter-sequence SIMD kernel that is bit-identical to a scalar Gotoh across x86-64, AArch64 and WebAssembly (Section 6);

5. an evaluation on two independent simulators with paired statistics, a negative-control suite, ablations of the D priors, the per-read call, the open-flank cost and the plausibility constraints, and an attribution of the remaining D errors to search or model (Sections 7 and 8).

2 Model

2.1 Problem

Given a read q and germline databases of V, D, J and optionally C alleles, grouped into genes, swig reports the strand of q, the alleles of its V, D (zero or one) and J segments with their read and germline coordinates, hence the N/P regions and the junction, and optionally the constant region. The read may carry substitutions and short indels from SHM, sequencing errors, trimmed segment ends, non-templated junction bases and unrelated flanking sequence such as primers, adapters or UMIs.

2.2 Segments are local alignments scored as log-odds

A *segment hypothesis*, an allele a and a read interval I, is scored by affine-gap local alignment [6, 7] of $q[I]$ to a. With match score m and mismatch score μ a substitution column scores $s(x, y) = m$

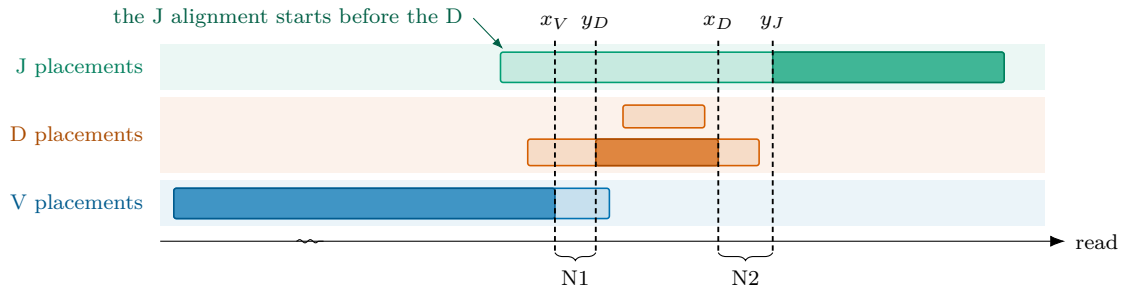


Figure 1: Placements and cuts (schematic; the junction is drawn wider than to scale). Pale bars are candidate local alignments (placements) of V, D and J alleles to one read; saturated bars are the parts a path keeps. Between consecutive segments the earlier one exits at x, the later one enters at $y \geq x$, and the bases between are N/P. Overlaps can be split anywhere or left to N/P. Because nodes are ordered by segment kind, not by read start, the path V→D→J is available even though the J alignment reaches back over the D. The placement graph maximises the objective of Equation (3) over all such paths, alleles and cut pairs.

if $x = y$, μ if $x \neq y$, and 0 if either base is ambiguous. These are log-likelihood ratios [8, 9]: let λ solve

$$\tfrac{1}{4}e^{\lambda m} + \tfrac{3}{4}e^{\lambda \mu} = 1, \tag{1}$$

the Karlin–Altschul λ for uniform base composition. Then $e^{\lambda s(x,y)}$ is the likelihood ratio of "x was templated by y at divergence δ" against "x is random", with $1 - \delta = \frac{1}{4}e^{\lambda m}$. For the default $(m, \mu) = (2, -3)$, $\lambda = 0.634$ nats per score unit and $\delta = 11.2\,\%$: the familiar $+2/-3$ nucleotide scores are the DNA analogue of a BLOSUM matrix [10] at that divergence. Gap runs of length L cost $g_o + (L-1)g_e$, with $(g_o, g_e) = (-9, -1)$ for V, $(-13, -1)$ for D and $(-17, -2)$ for J. N/P and flank bases score 0, since they are random under both hypotheses.

V and J alignments are extended co-linearly, without gaps, towards the germline's outer end (V 5′, J 3′) or the read's end, whichever comes first. The extension stops before the column at which its running score falls more than 30 below the alignment's edge (an X-drop [11] of 10 net mismatches), so an adapter or a random 5′ flank is cut after about 17 bases on average (after 10 when every base mismatches, later when chance matches occur) instead of being taken whole for V. Each stage keeps its best alignments both by extended and by local score, so that a hit at the read's end, which pays for no flank, cannot take every slot from a locally stronger hit that pays for a mutated one.

2.3 Placements, cuts and paths

A *placement* is an aligned segment hypothesis. A *path* is a sequence of placements allowed by the grammar

$$V \;\rightarrow\; D^{\{0,1\}} \;\rightarrow\; J, \tag{2}$$

with partial paths V, J, VJ, VD and DJ allowed (never a lone D). Between two consecutive placements there are two *cuts*: the first keeps the read bases before its *exit* x, the second keeps the bases from its *entry* $y \geq x$ on, and the $y - x$ bases between them are N/P. Each placement is *clipped* to the alignment columns it keeps. Overlapping placements may therefore split an overlap at any point, or leave part of it to neither. Only the cuts order segments of different kinds: a D may precede a J whose alignment starts before the D's, as long as the J is entered after the D's exit (Figure 1). This is precisely the configuration in which a sequential annotator lets a J alignment "eat" the D.

2.4 Objective

The score of a path is

$$\begin{aligned}\text{score}(\pi) \;=\; &\sum_{\text{segments}} S_{\text{clip}}(\text{segment}) \;-\; P_V\,\#V \;-\; P_D\,\#D \;-\; P_J\,\#J \\ &-\; c_{\text{open}}(\text{first}) \;-\; c_{\text{open}}(\text{last}) \;+\; \sum_D (b\,|D| - \tau_5\,t_5 - \tau_3\,t_3 + \kappa),\end{aligned} \tag{3}$$

where S_{clip} is the score of the kept columns. The other terms are log prior odds: P_V, P_D and P_J in score units, c_{open} and the D terms in nats, read at the call's scale (the fixed scale of Section 3.3 converts c_{open} to score units and has no D trim or N/P terms). Each has a stated derivation; of the constants it leaves free, the open-flank allowance and the credit κ were set on simulated reads, while the trim and N/P means are generic rather than fitted (Section 9).

Segment presence. A D hypothesis is one of N D genes placed at one of L junction positions, so its prior is spread over NL alternatives. A maximum a posteriori decision includes the best D when $\lambda S_D > \ln(NL) - \ln \pi_D$, with π_D the prior odds of a D against none, which gives

$$P_D^* \;=\; \frac{\ln(NL) - \ln \pi_D}{\lambda}. \tag{4}$$

Genes are counted after identical sequences are merged. For the IMGT human IGH set of Section 7 ($N = 30$, $L = 25$, even odds) $P_D^* = 10.45$: an exact 5-mer (score 10) loses and an exact 6-mer (12) wins. For TRB ($N = 2$) $P_D^* = 6.17$. V and J pay the same kind of cost over their own genes ($P_J^* = 7.91$ for its 6 IGHJ genes, $P_V^* = 12.54$ for its 113 IGHV genes). Without P_J and P_V, any positive clip completes or starts a path for free, so a read that stops before its J or starts after its V receives a J or V from a few chance bases. The cost is kept unrounded; on the integer fixed scale charging $\lfloor P_D^* \rfloor$ decides $S > P_D^*$ exactly. A database of several loci gives each locus its own costs. An alternative derivation counts the hypotheses a local D search actually tries, every allele offset and read position, through the Karlin–Altschul expectation $Kwn_De^{-\lambda S}$ (w read positions, n_D allele offsets). It is available as an option (`--d-presence-rate`) but needs a D-presence rate that does not follow from the search: at the D-presence rate of the IgSim *mid* reads (94.74 % carry a D remnant) P_D^* falls to 9.86, below an exact 5-mer's score of 10, and exact 5-mers win.

D trims and N/P lengths. A D clip that keeps germline $[r_{\text{lo}}, r_{\text{hi}})$ of an allele of length L_a trims $t_5 = r_{\text{lo}}$ bases at its 5′ end and $t_3 = L_a - r_{\text{hi}}$ at its 3′ end. Under geometric trims of means 5 and 7 each trimmed base costs $\tau = \ln((\bar{t}+1)/\bar{t})$, i.e. $\tau_5 = \ln(6/5) = 0.18$ and $\tau_3 = \ln(8/7) = 0.13$ nats. Under geometric N/P lengths of mean 8 a junction pays $\ln(9/8)$ per N/P base, so every read base a D keeps is one N/P base fewer and earns $b = \ln(9/8) = 0.12$ nats. The D cost is credited with $\kappa = 5\tau_5 + 7\tau_3 = 1.85$ nats, the trim cost of a D with the prior's mean trims. The trims then re-rank Ds while a typical D is called as before, and a short exact match deep inside a long allele, which used to be worth as much as an untrimmed short allele, pays for the germline it leaves out. Every term depends on a single cut, through its germline or read position, so a clip remains worth $\text{exit}(x) - \text{entry}(y)$ and the exact recursion of Section 3 is unchanged.

Open flank. A path whose last segment is a D asserts a junction whose J the read does not show in the x bases after the D; one whose first segment is a D, a V missed in the x bases before it. That D pays the log-likelihood ratio of those bases under "no junction, or a junction whose J (V) was missed" against "a junction whose J (V) was missed":

$$\begin{aligned} c(x) &= \ln\Big(1 + \rho\,\frac{1 - \varepsilon_{\text{eff}}(x)}{\varepsilon_{\text{eff}}(x)}\Big), \qquad \varepsilon_{\text{eff}}(x) = \varepsilon\big(\min(x - N_{\text{open}},\, 64)\big), \\ \varepsilon(\ell) &= \max\Big(\varepsilon_{\min},\; \min_{1 \le \ell' \le \ell} \Pr\big[X_{\ell'} \ge k(\ell')\big]\Big), \qquad X_\ell \sim \text{BetaBinomial}(\ell,\, \alpha,\, \beta), \end{aligned} \tag{5}$$

where $k(\ell)$ is the number of mismatches among ℓ visible bases that keeps them below the smallest score the missing segment's stage would accept ($\varepsilon = 1$ for $\ell \leq 0$ and while no ℓ bases can reach that score), and $\mathrm{Beta}(\alpha, \beta)$ is the read's mismatch rate from its anchored V and J columns ($\alpha = k_a + \frac{1}{2}$, $\beta = n_a - k_a + \frac{1}{2}$ for k_a mismatches among n_a columns). $\varepsilon(\ell)$ is thus the probability that a segment present in the read would have stayed below the detection threshold, and the cost is constant beyond 64 visible bases. The free allowance $N_{\mathrm{open}} = 24$ is the largest 99.9th percentile (22–24 across simulated sets) of the distance from a called D to the true J start, or from the V end to the D; $\rho = \frac{1}{2}$ (even odds) and $\varepsilon_{\min} = 10^{-3}$ (the anchor E-value gate) bound the cost to [0, 6.2] nats. At high divergence a missing segment could well have been missed and the cost falls accordingly: with 13 mismatches among 45 anchored columns (about 30 % SHM), 64 visible bases cost 0.99 nats for a missing J and 0.71 for a missing V, against 6.2 on a clean read. A real D lies within its N/P region of the read end, a chance D anywhere in the search window, so the cost removes far chance Ds while keeping real Ds of reads that stop at their D or start in CDR3 (Section 8.4).

Order of the decision. The adjudicator returns the path that maximises, lexicographically, (1) its value, by default the read's own log-odds (Section 3.3); (2) fewer D segments, so a path with a D must strictly beat the best path without one; and (3) germline *support*, the sum of its placements' unclipped scores. Key (3) separates paths that explain the read equally well, typically two alleles of different 3′ length at a junction base that can belong to either side; the allele with more support admits more cuts with the same score and hence has the larger likelihood summed over cuts. Co-optimal alleles, equal in clipped score and in matches and mismatches on the called interval, are all reported. J names are positional: a tied J allele is named only when it explains the called read bases at the first allele's own germline coordinates (the same germline interval), so IGHJ4*02 and IGHJ5*02 trimmed to their shared 3′ part, which tie from different germline offsets, are not both named; the other remains a tied alternative counted by the posterior.

2.5 Significance

Each V and J hit receives a Karlin–Altschul E-value [9] $E = Kwne^{-\lambda S}$ against its database, with w the length of the searched read window and n the database length, using exact ungapped λ and K for the scoring in use. A hit is kept only at $E \leq 1$, and a read is *called* only when its path contains an *anchor*: a V or J hit at $E \leq 10^{-3}$ whose clip on that path still reaches the anchor score. Chance calls on random sequence are therefore rare, of order 10^{-3} per V or J search under the ungapped statistics; none of 2,000 random reads was called (Section 8.3). A J downstream of a significant V needs only $E \leq 1$, since given the V a J is expected there.

3 Exact adjudication

3.1 Clipping and separability

Lemma 1 (clipping never helps)**.** *Let A be an optimal local alignment and $A[a..b]$ a contiguous sub-alignment that begins and ends in match columns. Then $S(A[a..b]) \leq S(A)$.*

Proof. Every prefix and suffix of an optimal local alignment has non-negative score, or removing it would give a higher-scoring local alignment. Hence $S(A[a..b]) = S(A) - S(\text{prefix}) - S(\text{suffix}) \leq S(A)$. □

A placement is flattened into columns with prefix scores $P[k]$ and two tables mapping each read offset to the first match column at or after it and the last match column before it. A clip to $[lo, hi)$ is then two lookups and one subtraction, $S = P[\text{exit}(hi)] - P[\text{entry}(lo)]$, and it is exact

because a clip starts and ends on match columns and so never splits a gap run. The clipped score has one term per cut, *separability*, and the support key does not depend on the cuts.

3.2 The recursion

Placements are the nodes of a lattice, sorted by segment kind (V, D, J) and then by read start. Every transition between kinds goes forward in this order wherever the two alignments start, and the cuts alone decide whether it is feasible ($x \le y$). The paths through a node with a given D count and first segment kind form a *slot*, which holds one value per canonical entry cut y of the node: the best key of a path prefix entering there. A path's first segment keeps its start and its last segment its end, since V and J are extended outward on purpose; a path can therefore start at a node only at the node's own start, in the slot whose first kind is the node's. With a linear N/P score $\iota(y - x)$ ($\iota = 0$ by default) the recursion is

$$X(b, x) = \mathrm{exit}_b(x) - P_{k(b)} + \max_{y \le x - m_b} \big[F(b, y) - \mathrm{entry}_b(y)\big], \tag{6}$$

$$F(e, y') = \max_b \max_{x \le y'} \big[X(b, x) + \iota\,(y' - x)\big] \quad (-c_{\mathrm{open}}(e) \text{ at } y' = \mathrm{start}(e) \text{ for a path starting at } e), \tag{7}$$

$$\mathrm{best} = \max_b \; X\big(b, \mathrm{end}(b)\big) + \mathrm{end}(g, b) - c_{\mathrm{open}}(b), \tag{8}$$

over every exit x of the predecessor b and every entry $y' \ge x$ of the successor e, where $P_{k(b)}$ is the presence cost of b's kind, m_b its minimum number of retained bases, and $\mathrm{end}(g, b)$ the whole-path terms of b's slot g (such as an optional V–J pair bonus, 0 by default). Because each term depends on one cut:

1. the best entry for an exit, $\max_{y \le x - m_b}[F(b, y) - \mathrm{entry}_b(y)]$, is a running maximum along the node;
2. the best exit for an entry, $\max_{x \le y'}[X(b, x) - \iota x]$, is a second running maximum along b, built once per slot of b and per successor kind, not per edge;
3. every entry $y' \ge \mathrm{end}(b)$ sees the same best exit of b, so the edge $b \to e$ adds it once at e's first entry at or after $\mathrm{end}(b)$, and a prefix maximum over e's entries carries it to the rest.

An edge therefore costs $O(\mathrm{overlap} + 1)$ and a read $O\big(\sum_{\mathrm{slots}} \mathrm{len} + \sum_{\mathrm{edges}} (\mathrm{overlap} + 1)\big)$, plus one scan of each node's predecessor slots; Algorithm 1 gives the recursion as pseudocode. With the open-flank cost on, one transition is barred when the lattice opens its slots: a V or J none of whose clips pays its presence cost may not precede or follow a D, since such a placement is no evidence of the junction's V or J (a junk J of a base or two would otherwise close a D's flank for less than the cost). The open-flank cost is a constant of the node and the read added once where a path starts and once where it ends, and the D priors are added to a D node's entry and exit prefix values once, when the node is laid out; neither breaks separability. The recursion is a semi-Markov Viterbi [12, 13]: segments of variable length scored as wholes, with the cut positions as the hidden boundaries.

Theorem 1. *The recursion of Equations (6) to (8) returns the optimum of Equation (3), with the tie order of Section 2.4, over all grammatical paths (with the transition rule above when the open-flank cost is on), all placements and all pairs of canonical cuts.*

Proof sketch. Every grammatical predecessor of a node precedes it in the order by kind, so by induction over that order $F(e, y)$ is the best value of a path prefix that enters e at y, and $X(b, x)$ that of a prefix that leaves b at x. A path's value is a sum of terms each of which depends on a single cut, so the maximum over the pair (x, y) of an edge factorises into the two running maxima above, and the maximum over the entries beyond b's end into one shifted prefix maximum. Ties are resolved by comparing the lexicographic key (value, fewer D, support), each of whose parts is separable in the same way. □

Algorithm 1 Exact adjudication of one read. The posterior pass runs the same loop in the sum-product semiring, with prefix sums in place of running maxima.

Require: placements of the read sorted by kind (V, D, J), then read start; for each placement b the prefix values $\text{entry}_b(y)$ and $\text{exit}_b(x)$ at its canonical cuts, D priors included

for each node e in order, and each slot of e (D count, first kind) **do**
  $F(e, y) \leftarrow -\infty$ for every entry y
  if this is e's start slot and e may start a path **then**
    $F(e, \text{start}(e)) \leftarrow -c_{\text{open}}(e)$
  end if
  for each predecessor b whose slot and kind may precede e **do**
    $F(e, y) \leftarrow \max\big(F(e, y),\, M_b(y) + \iota y\big)$ for every entry $y < \text{end}(b)$
    offer $M_b(\text{end}(b)) + \iota y_0$ at the first entry $y_0 \geq \text{end}(b)$
  end for
  carry the offers to later entries by a prefix maximum, $+\iota$ per base
  $G(x) \leftarrow \max_{y \leq x - m_e} \big[F(e, y) - \text{entry}_e(y)\big]$ ▷ running maximum along e
  $X(e, x) \leftarrow \text{exit}_e(x) - P_{k(e)} + G(x)$ for every exit x
  $M_e(x) \leftarrow \max_{x' \leq x} \big[X(e, x') - \iota x'\big]$, once per kind that may follow e
end for
return $\max_b X(b, \text{end}(b)) + \text{end}(g, b) - c_{\text{open}}(b)$ and its trace-back

The search runs over every canonical entry of the successor, not only those inside the overlap. Restricting entries to the overlap appears justified by Lemma 1, but the lemma bounds prefixes that start at the alignment's own start, whereas the bases between b's end and a later entry are a middle stretch of e's alignment and can score below zero: two mismatches right after the V's end, as a mutated read has, are better left to N/P. Likewise, ordering nodes by read start alone excludes a D before a J whose alignment starts earlier. Both restrictions were found by adversarial review of earlier versions; swig's test suite now checks Theorem 1 against a memoised exhaustive enumeration of every grammatical path and canonical cut pair, on synthetic reads, on the pipeline's own placements for hypermutated reads, under a two-D grammar (VDDJ optima occur), with the per-read model at five divergences, with the open-flank cost and with the D priors, and on constructed reads that each restricted search gets wrong.

3.3 Calls at the read's own divergence

The scores $(2, -3)$ are log-odds only at 11.2 % divergence. A fixed scale makes unmutated reads under-confident, since a single nucleotide difference between two alleles is worth $\ln(3(1-p)/p) = 8.0$ nats at $p = 0.1\,\%$ rather than $\lambda \cdot 5 = 3.2$, and heavily mutated reads over-confident. swig therefore estimates the read's divergence $\hat{p}$ before adjudication from the columns no path contests—the best V's match columns before every D and J placement and the best J's after every V and D placement, N columns excluded, or all columns of the two placements when fewer than 50 remain—as the Jeffreys predictive rate $(k + \frac{1}{2})/(n+1)$ for k mismatches among n columns [14], clamped to $[10^{-4}, 0.5]$. Apart from that fallback the junction bases are not among these columns, so they are not used twice. A substitution column of score $s \neq 0$ is then worth $\lambda_r s + \mu_r$ nats, with (λ_r, μ_r) chosen so that a match is worth $\ln 4(1-\hat{p})$ and a mismatch $\ln(4\hat{p}/3)$; at $\hat{p} = \delta$, $\lambda_r = \lambda$ and $\mu_r = 0$. Gaps, ι and the pair bonus stay at the prior λ, since $\hat{p}$ measures substitutions.

Every term is an integer in units of $1/F$ nats, $F = 2^{16}$: a column of kind $k \in \{V, D, J\}$ is $s\,u_k + v_k$ with $u_k = \text{round}(F\lambda_r)$ and $v_k = \text{round}(F\mu_r)$, (λ_r, μ_r) computed from that kind's own match and mismatch scores (identical for the default $(2, -3)$), so the maximum and every tie are exact, and the recursion and Theorem 1 carry over unchanged. At $\hat{p} = \delta$ every column is the fixed score times $u_0 = \text{round}(F\lambda)$, so with integer presence costs, the D priors off and the open-flank cost off, the fixed scale (`--calls fixed`, which has no D trim or N/P terms and charges the open-flank cost in whole score units as $\lfloor P_D^* + c_{\text{open}}/\lambda \rfloor - \lfloor P_D^* \rfloor$) is an exact special case. Each presence cost is read as P_k^*/m matched columns of its kind rather than as a constant

number of nats, with m the match score (2 for every kind): $c_V(\hat{p}) = \lfloor P_V^*(mu_V + v_V)/m \rfloor$ and $c_J(\hat{p}) = \lfloor P_J^*(mu_J + v_J)/m \rfloor$. A matched D column also earns the N/P term round(Fb), and the D cost is lowered by the credit:

$$c_D(\hat{p}) = \left\lfloor P_D^* \, \frac{mu_D + v_D + \mathrm{round}(Fb)}{m} \right\rfloor - \mathrm{round}(F\kappa). \tag{9}$$

The D cost is a threshold against chance D hits in random N/P sequence, whose null distribution does not depend on $\hat{p}$ and is dominated by exact runs. Read in matched columns it keeps the fixed scale's admitted classes of exact runs at every divergence for a D with the prior's mean trims (runs of 5 lose, runs of 6 win when their trims cost κ); the trim terms move runs that are trimmed less or more than that across the threshold. A cost constant in nats would instead admit exact 5-mers with the prior's mean trims on unmutated reads.

3.4 Posterior probabilities

The objective is a log-likelihood ratio, so it also defines a distribution over every grammatical path and every pair of canonical cuts,

$$P(\pi, \mathrm{cuts} \mid q) = \frac{\exp\big(\mathrm{nats}(\pi, \mathrm{cuts})\big)}{Z}. \tag{10}$$

Forward–backward [15, 16] over the same lattice—the sum-product twin of Equations (6) to (8), with prefix and suffix sums where the Viterbi pass has running maxima—gives the marginal of every placement and of every D count in the same $O(\mathrm{overlap} + 1)$ per edge. Under the default call both passes read the same integer model, so the Viterbi path is a mode of the reported distribution. A segment's reported posterior is the mass of every placement of its kind that is tied with the call on the called interval or explains most of the called bases with germline the primary allele contains; the event is "the reported names include the read's allele, up to germline equivalence", which is exactly what the evaluation scores (Section 7). The read's D-presence probability is $1 - P(\text{no D})$.

The arithmetic is linear with one log-scale per slot, as in scaled HMM recursions. Cut weights are running products along the alignment, so no transcendental function is evaluated per cut; entry and exit weights are centred separately; the N/P factor is accumulated as a relative recursion whose factors are ≤ 1; and the cost of a gap between non-overlapping neighbours joins the slot's log-scale rather than multiplying it. A read whose evidence would leave the range of double precision (a placement spanning more than 1,200 nats on one side, or an overlap gaining more than 600) receives no posterior rather than a wrong one. Tests compare the marginals with exhaustive summation under VJ, VDJ and VDDJ grammars, with N/P costs, N columns, insertions, long costly gaps, the open-flank cost and the D priors.

An optional decoder (off by default) takes a call with exactly one D and finds the allele of largest posterior marginal, $a^* = \arg\max_a \sum_{p \ni a} P(p)$ over D placements p; the marginal is not conditioned on the call's topology. When a^* beats the call's allele, the path is re-traced with the call's first and last kind and D count on the lattice masked to the V and J placements and a^*'s D placements, and kept when it stays anchored and its reported D posterior is no lower.

4 Plausibility constraints

On full-length simulated reads the optimum of Equation (3) almost always pairs a V and a J that could form one rearrangement, but on truncated, chimeric and cross-locus reads it need not. swig applies, as constraints outside the objective and on by default, the plausibility checks of SwiftIG, an earlier C++ aligner (unpublished) whose method note swig began as a Rust implementation of: minimum hit lengths of 24 (V) and 10 (J) local alignment columns; a V–J

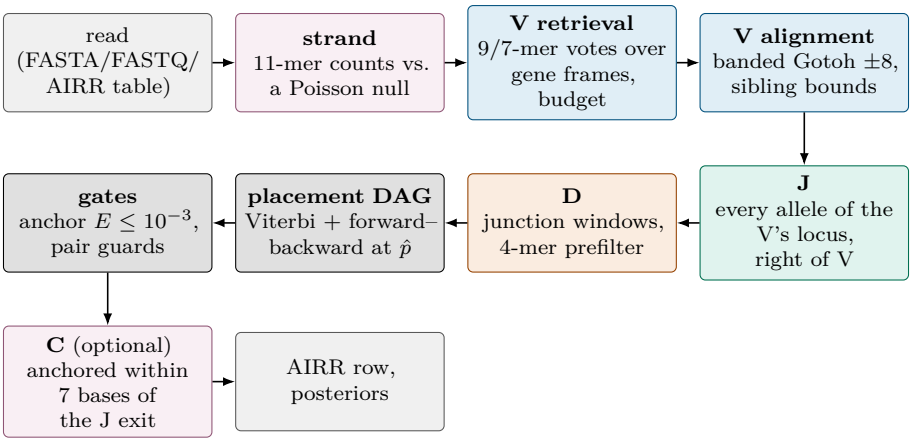


Figure 2: Pipeline of one read. The V, J, D and C alignment stages run over chunks of 64 reads on the inter-sequence SIMD kernel (16 or 32 lanes); strand, windows, admission, adjudication and the gates are decided per read. Table 1 gives the guarantee of each step.

pairing rule (the V and J align both conserved codons in order, or hold at least 80 local alignment columns together and either face their junction or hold a long split), to which swig adds that, unless the pair aligns both conserved codons in order, a V shorter than 80 columns pairs only with a J that is significant on its own; and a span of at most 225 bases between the V's exit and the J's entry. SwiftIG called a D only between a V and a J that pair. swig also admits a D beside a V or a J alone, but only when it pays the open-flank cost and lies within 60 bases of the segment the read shows. The call is the unconstrained optimum when it satisfies the constraints; otherwise the lattice is re-adjudicated over subsets of placements (the V placements alone, the J placements alone, and each J with the V placements it pairs with, with and without the Ds; under the open-flank cost also V and D placements within reach of each other, and D and J placements), and the call is the best admissible subset optimum. Within each subset the key is the exhaustive optimum, but a subset whose optimum breaks a constraint contributes nothing, so the guarded call is *defined* rather than exact. On the test suite's 435 checked calls, no admissible path over any choice of at most one V, D and J among a read's best 4, 8 and 4 placements beat the call under the per-read model (one did under the fixed scale). Junctions are reported only from conserved anchors (Cys104, Trp/Phe118) that the called alignments actually align, with a bounded recovery of a Cys104 lying just past a clipped V end.

5 Search

Each stage narrows the hypotheses the lattice sees. Table 1 states what each step can lose: *exact*; *lossless*, which provably drops nothing the model could choose; *bounded*, which drops only what lies outside a stated geometric bound; *defined*, a stated selection rule whose output is the definition of the result rather than an approximation of an optimum; *approximate*, a known gap to the model's optimum; and *heuristic*, which is measured, not proven. Figure 2 shows the flow.

Gene frames and the index. The alleles of a gene share almost all seeds. An allele joins its gene's coordinate frame when at least 30 % of its seeds occur in the gene's longest allele and every one of those shared seeds agrees on one offset; SNP and truncated alleles join, an allele with an internal indel gets its own frame. The k-mer index is built over frames in compressed-sparse-row form, with 2-bit packed keys used directly as array indices, so a seed shared by twenty alleles is one posting and one vote and retrieval work scales with genes rather than alleles. Votes are packed into 64-bit integers and radix-sorted in linear time; one scan then yields the best diagonal cluster per frame. Seeds with more than 256 postings are skipped, and a seed repeated within a

frame does not vote, so repeats cannot collect quadratic votes. We measured closed and open syncmers and minimisers [17, 18] as seed samplers (minimisers only under an earlier fixed gene margin). Under the noise-based margin every k-mer beat closed syncmers by 1.1 points of mean V gene accuracy (4.4 at 30 % SHM) for 0.4 % more instructions, and beat open syncmers too, so it is the default.

Sibling bounds. For a sibling A aligned in the same frame diagonals as its representative R, $\Delta(R, A)$ sums the best profile gain at each frame position both alleles cover and the best substitution score where only A does, so $S^*(A) \leq S^*(R) + \Delta(R, A)$. The lower bound $\underline{L}$ rescores a traced representative path clipped to A's span with A's profile, in time proportional to the positions where the two differ. A sibling whose bounds meet needs no alignment, and one whose lower bound already reaches the cut-off is aligned outright with its traceback reused. Because the cut-off is a lower bound on the true third score, the kept ranking is that of aligning every allele.

D search. The junction windows reclaim 12 bases into V and J, and the conserved-anchor window depends on where the J's Trp/Phe118 lands, not on how far its alignment reaches left; with the lattice's ordering by kind, the read's own J cannot remove its D from consideration. A chance J placement between the V and the D of a read that stops at its D can, since the window ends 12 bases after the start of the best J placement, used or not. Each D allele sharing an ℓ-mer with a window is aligned there, unbanded, and its optimal local alignment competes for the 32 places (plus ties) that the window's admissible D hits have in the lattice. D admission requires an exact run of at least 4 bases; 3, 5 and 6 were measured and are worse.

5.1 Constant region

With a C database the constant region is assigned after adjudication, from finished calls only, so V, D and J calls are unchanged by construction. J and C are joined by an RNA splice at fixed phase, so C's germline base 0 lies at the called J's exit x, up to reference conventions and indels near the J's 3′ end; hypothesis $H(a, o)$ places base 0 of allele a at $x + o$, $|o| \leq 7$. An alignment anchored at $C[0]$ on one of these 15 diagonals is computed exactly by any local aligner run on $[N] + C$ whose first column scores a bonus larger than any alignment avoiding it, so the SIMD kernel serves unchanged. Alleles whose sequences agree on the reachable prefix form reach classes and are aligned once (the IMGT human IGHC set used here has 105 distinct alleles but only 13 distinct 32-nt prefixes). C is called when its anchored E-value $E_C = 15\, N_C\, e^{-\lambda S} \leq 10^{-3}$, with N_C the number of C genes of the J's locus, from a Lundberg-type bound for the walk from each start, on at least 12 aligned columns at 90 % identity, and its posterior is computed at C's own divergence, since C is not hypermutated.

5.2 Germline annotation

Conserved anchors fix every reading frame and junction end: V Cys104 and J Trp/Phe118. They and the V region boundaries are settled once per database, in order of trust: header keys, IMGT-gapped numbering when the gaps are consistent, a joint motif search for Cys23, Trp41 and Cys104, transfer from the most similar allele by local alignment, and finally database models: a Jeffreys-smoothed model of the FR3 length given the Trp41–Cys104 spacing, and of the Cys23 position. Every derived value that may be wrong carries an *uncertain* flag, so an error without the flag is a silent error. Each allele's locus is read from its name or inferred from its relatives and, failing that, from an embedded panel of human and mouse family representatives; the locus decides which alleles may form one rearrangement and which gene counts price a segment. V alleles named without the ∗ allele separator join the gene of their nearest named relative (≥ 0.95 of the smaller self-score; the rest are grouped among themselves), so the gene-level retrieval budget does not silently become an allele cap.

6 Implementation

swig is a Rust crate (edition 2024, Rust $\geq$ 1.89) with no dependencies: about 38,000 lines of library code with inline unit tests, 5,400 lines of integration tests and 1,300 of examples and WebAssembly bindings. Every pipeline component is a statically dispatched trait—aligner, retrievers, candidate budget, D window policy, admission, adjudicator, grammar and path model—so swapping one changes the type and the pipeline monomorphises without trait objects. The substitution model is also a trait, applied once when the database is prepared to build the per-allele profiles that the kernels read as data.

Inter-sequence SIMD. Following Rognes [19], each 16-bit lane runs the scalar Gotoh recurrence for a different read–allele job rather than striping one alignment [20]. The kernel is written once over a lane abstraction and compiled for AVX-512BW (32 lanes), AVX2, NEON, WebAssembly [21] `simd128` and portable lanes (16). By default the fastest of AVX2, NEON and portable lanes is chosen at run time; AVX-512BW runs only on request (`--simd avx512`), since 512-bit instructions lower the clock of many CPUs. On WebAssembly the backend is fixed at build time (`simd128`, or the scalar kernel without it). Banded jobs use band-relative coordinates, so lanes whose bands start on different diagonals walk the same grid [22]. When every lane's profile depends on base identity alone a cell's substitution is one comparison and one blend; otherwise five profile rows are selected by mask, exact for any substitution model. The horizontal gap state uses $D' = \max(t + g_o,\ D + \max(g_o, g_e))$ with $t = \max(0, \mathrm{diag}, I)$, which shortens the loop-carried dependency to one add and one maximum and is exact for any gap costs. Arithmetic saturates in 16 bits; a job enters a lane only if its attainable local score is at most 30,000, no step costs less than $-1{,}000$ and its dimensions fit 16-bit positions, so saturation never clips a real cell, and other jobs run on the scalar kernel. Traceback codes are bit-identical to the scalar kernel's, so one traceback walk serves both.

Batching and determinism. Each stage runs over a chunk of 64 reads, giving one score batch and a few traceback waves per stage, while every decision stays per read; a read's result does not depend on its chunk, which is tested. Threads self-schedule chunks from an atomic counter with one workspace each, so there is no allocation in the DP after warm-up, and results return in input order. The scalar kernel equals a textbook Gotoh, every SIMD backend equals the scalar kernel in scores, end cells and tracebacks, the batched pipeline equals the per-read scalar pipeline, and output is byte-identical across backends and between x86-64 and AArch64.

Input, output and interfaces. Reads are FASTA, FASTQ or an AIRR Rearrangement table, streamed in batches of 16,384; identical sequences are assigned once per batch, and across batches through a cache of the 4,096 most recently assigned distinct sequences, which changes no output. Output is the AIRR Rearrangement schema [23] with the posteriors, E-values, alternatives and region coordinates. The crate provides a command line (`swig`, with `prepare`, `simulate` and `evaluate` subcommands) and a Rust library; a companion workspace crate (`swig_wasm`) builds a WebAssembly module with a C ABI. All three share one run implementation.

Tests. The default suite of 350 tests, all passing on the version described (one further long-running accuracy test, of 20,000 reads per preset, runs on demand and is not counted among them), includes the exhaustive oracles for Theorem 1 and the posterior marginals, kernel equivalence on random batches for every backend the host runs (AVX-512BW, AVX2 and portable lanes on x86-64; NEON and `simd128` on AArch64 and WebAssembly builds), the guards on constructed reads, robustness on degenerate input (empty, all-N, IUPAC, low-complexity, 20 kb and chimeric reads, and every prefix and suffix of a read), and accuracy floors on simulated reads. It runs on release code with debug assertions and overflow checks enabled.

7 Evaluation

All reads in this paper are simulated, so that every call can be scored against a known answer; none of the results is external biological validation (Section 9). Two independent simulators are used. The numbers in Section 8 were produced with swig 0.39.7 for this paper, except the MiXCR results, the analysis of the previous release's D errors and the D-prior prototype measurements of Section 8.1, which come from earlier runs of the GenAIRR benchmark on the same reads, and the instruction count in that section, from the 0.39.7 release measurements (callgrind on 1,000 IGH reads).

GenAIRR. GenAIRR 2.3.1 [5] simulates reads with its built-in references (human IGH from OGRDB [24]: 198 V, 33 D and 7 J alleles; IGK 168 V and 8 J; IGL 181 V and 10 J; TRB 98 V, 3 D and 16 J), 10,000 reads per set, seed 7, forward strand. Scenarios are unmutated (*naive*); S5F mutation [25] at 2, 5 and 10 %; and *realistic*: S5F at 5 % with a 5′ primer trim of 0–15 and a 3′ trim of 0–5 bases, 0–1 polymerase indels, 0.2 % PCR and 0.2 % sequencing errors and 0–2 N calls (for TRB, the same artefacts without SHM). Every tool receives the GenAIRR reference: swig with each allele's anchor in its header, IgBLAST as `makeblastdb` databases with a J auxiliary file, MiXCR as a repseqio library. GenAIRR's truth for each segment is a *tie set*: each allele earns a point for every read base that matches it at the same germline index as the sampled allele, over the segment's true read region, and every allele at the maximum is in the set. A call C against truth T scores the Jaccard index $|T \cap C|/|T \cup C|$ per read and segment (both empty scores 1, one empty 0), and tables give the mean in %. One correct allele against a two-allele tie therefore scores $\frac{1}{2}$, as does a two-allele call against a one-allele truth.

IgSim. swig ships a dependency-free Rust port of the generative model and presets of IgSim.jl [26]: uniform allele choice, exonuclease trims, N additions, random flanks, gated SHM with single-base indels, and instrument substitutions, with every read base labelled by its germline origin. The presets are *easy* (no SHM), *mid* (IgM-like: SHM on 40 % of reads at 0.5–7 %; 1-nt indels switched on for 8 % of reads at up to 0.2 % per base, so about 2 % carry one) and *hard* (IgG-like: SHM on 97 % of reads, median about 7 %; indels on 15 % at up to 0.4 % per base, about 7 % carry one); we also run the *hard* preset with the SHM rate fixed at 30 % on every read to probe the extreme. The germline is the human IGH set of IMGT/GENE-DB [27] release 202632-7 (480 V, 48 D and 15 J alleles, of which swig merges identical sequences to 457, 41 and 15), given to every tool as plain FASTA: swig annotates the anchors itself. The simulator draws only V and J alleles with a trusted anchor, since a junction truth built on a guessed anchor would rest on the annotation it is meant to test. An allele call is correct when it is *germline-equivalent* to the truth: the called allele contains the trimmed, pre-mutation germline that was simulated, so no method could separate the two from the read. Junctions are correct when they equal the true Cys104–Trp/Phe118 sequence; junction accuracy is over the reads whose two anchor codons are templated in the read (9,953, 8,130 and 4,667 of 10,000 on *easy*, *mid* and *hard*; 1,441 of 3,000 at 30 % SHM). Because the trade-off between finding short D remnants and leaving D-free junctions without a D can be moved at will, D accuracy is always reported together with the share of D-free reads that are given a D, and by remnant length.

Comparisons and statistics. Two callers or settings are compared on identical reads. For each metric `swig evaluate`, which scores any AIRR Rearrangement table and so serves every tool alike, computes the accuracy difference with its 95 % paired Wald interval and an exact two-sided McNemar test [28] on the reads only one of them gets right; we report the differences and the McNemar p-values. Posteriors are scored by reliability bins, the expected calibration error [29] ECE $= \sum_b n_b\,|\bar{p}_b - \bar{e}_b|/n$ and the Brier score [30].

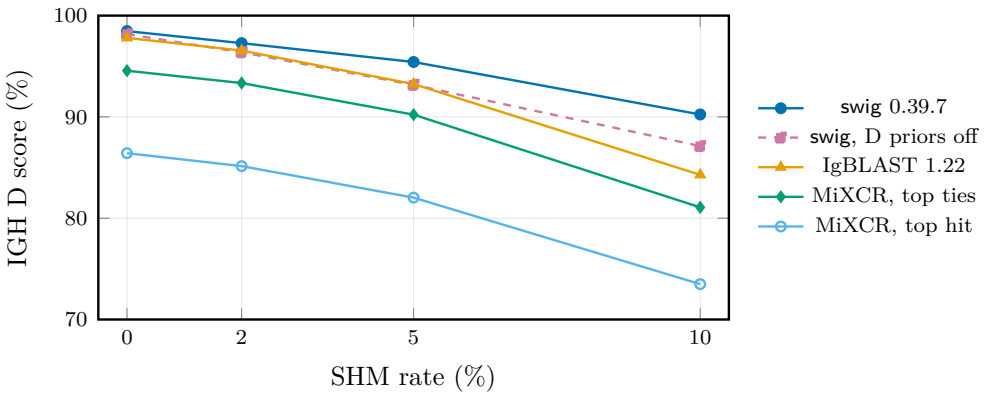


Figure 3: Human IGH D accuracy on GenAIRR reads as SHM increases (mean Jaccard score against the truth's tie set). "D priors off" is swig 0.39.7 with `--d-trim off --np-length off`, which reproduces the D calls of the previous release.

Tools and hardware. IgBLAST 1.22.0 (Bioconda) was run with the options `-organism human`, `-ig_seqtype Ig` (`TCR` for TRB) and `-outfmt 19`, with custom V, D and J databases built from the same FASTA. Its auxiliary file lacked IGHJ5*03 and IGHJ5*04 of the IMGT set; we added them with the frame and CDR3 end of IGHJ5*01, which swig's anchors reproduce. MiXCR 3.0.13, the last release that runs without a licence key, was run with `align` and `exportAlignments`, and is reported counting every hit tied with its top score, the fairer comparison with tools that list co-optimal alleles. Its figures come from an earlier run of the same GenAIRR benchmark on identical reads: the simulations are seeded, and our regenerated IgBLAST and swig scores on them reproduce that run's to the last reported digit. swig 0.39.7 ran with its default profile. All timings are wall-clock times of the whole command on a 4-vCPU Intel Xeon virtual machine at 2.1 GHz (AVX-512BW) with 15 GB of memory, on four threads unless stated.

8 Results

8.1 Accuracy against ambiguity-aware truth

Table 2 compares the three tools on GenAIRR reads. swig has the highest V, D and J score on every human IG set. The margin is largest for IGH D and grows with mutation (Figure 3): from 0.7 points over IgBLAST on unmutated reads to 6.0 points at 10 % SHM, and 2.4 points on the realistic set. MiXCR's top-tied hits trail on D by 3.9–9.2 points over the IGH sets; its single top hit, which drops genuine ties, scores lower still (IGH D 73.5–86.4 %, V 94.6–99.6 %). On TRB swig has the best D (99.9 and 99.5 % against at most 98.2 %) but trails IgBLAST and MiXCR by 0.04–0.11 points on J: on a few reads the J is trimmed 31–40 bases into its germline, past its Trp/Phe codon, and its 13–22 remaining bases fail the V–J pairing rule of the plausibility constraints (the J starts more than 30 bases into its germline, holds fewer than 30 columns and aligns no conserved codon).

Where the D gain comes from. Without its D trim and N/P-length priors swig 0.39.7 makes the D calls of its previous release, which was level with IgBLAST at 2 and 5 % SHM (Figure 3). An analysis of that release's errors on a 2,000-read sample of the 5 % SHM set found that 95 of its 103 wrong-gene D calls overlapped the true D: the D had been found, but a short exact match of another gene inside a mutated true D (for example `CTACAAT` inside `ACTACAATAA`) outscored it by local alignment. A post-processing prototype that held the placement graph's V end and J start fixed and re-decided the D under a posterior over allele, offset and both trims showed that the gain came from the trim and N-length priors and not from a different decision

rule or hypothesis space: with flat priors it matched the graph's D calls (93.12 against 93.16 at 5 % SHM), and restricting each allele to one offset, as the graph does, cost at most 0.1 point. Moving both priors into the graph as cut terms (Section 2.4) recovered most of the prototype's gain (95.42 against 95.67 at 5 % SHM) at 0.6 % more instructions, whereas the prototype as a separate pass would have added 37–47 % to the run time.

Because the priors were designed on these reads, we simulated four IGH scenarios again from a disjoint part of GenAIRR's random stream, sharing no read with the sets above (Table 3). The picture is unchanged: the priors add 0.5–3.4 points of D, at least as much as on the design sets, and swig leads IgBLAST on V, D and J in every scenario, on D by 0.9–6.3 points.

8.2 Germline-equivalent accuracy, D-free junctions and junctions

Table 4 compares swig and IgBLAST on IgSim reads against the IMGT human IGH set, read by read. swig calls the V allele more often correctly on unmutated, IgM-like and IgG-like reads ($p < 10^{-6}$ each), the D allele on unmutated and IgG-like reads (+2.2 points on *hard*, $p = 10^{-13}$), and the J allele on IgG-like reads, while giving a D to far fewer D-free junctions (18.4 against 49.4 % on *mid*, 15.8 against 30.8 % on *hard*). On remnants of 8 or more nucleotides swig's D is better on every preset. IgBLAST's 0.5-point higher D accuracy on *mid* ($p = 0.017$) comes entirely from remnants of 1–3 nucleotides (74.3 against 36.6 %), which no method can identify: almost any allele containing the remnant is germline-equivalent to it, so a tool that always calls a D is credited there, and pays for it on D-free junctions.

IgBLAST reports more exact junctions (99.94 against 99.53 % on *mid*, 98.33 against 96.81 % on *hard*). This is the cost of swig's plausibility constraints, which report a junction only from conserved anchors that the called V and J alignments actually align: without them swig's junctions are 99.98 and 99.66 % exact (Table 6), above IgBLAST's. The constraints in turn withhold junctions from reads whose anchor codon is not templated in the read (*hard*: 20.9 % of such reads receive one, against 98.0 % without the constraints and 98.5 % for IgBLAST).

At a fixed 30 % SHM, beyond IgSim's IgG-like regime, the picture is mixed. swig keeps a large lead on D (+7.3 points) and J (+13.6), but IgBLAST is better on V allele (+3.4) and V gene (95.0 against 89.7 %) and on junctions. swig leaves 4.4 % of these reads uncalled (133 of 3,000), 106 of them (3.5 % of all reads) because its strand test chose the wrong strand, as the strand test and seed retrieval run out of intact k-mers (Section 9); every read it calls is correctly oriented.

8.3 Specificity

swig made no call on 2,000 random reads with the length distribution of the *mid* set, and none on 1,000 IGK reads assigned against the IGH database: the anchor gate requires a V or J hit at $E \leq 10^{-3}$. IgBLAST, whose default E-value threshold for saving hits is a permissive 20, gave a V to 1,272 random reads (and a junction to 93) and an IGHV to 997 of the IGK reads. Filtering its V hits at $E \leq 10^{-3}$ removes every random-read call but keeps an IGHV on 470 of the 1,000 IGK reads.

8.4 Truncated reads and the open-flank cost

Real reads often stop at their D (short reads, 3′ fragments) or start inside CDR3, and a random tail beyond a V or before a J offers a D search window where no D exists. Table 5 cuts IgSim reads accordingly. With the open-flank cost, a D at the read's open end is called right on 88 % (*mid*) and 82 % (*hard*) of reads cut at the D's end and on 94 and 83 % of reads that start after the V, while a V followed by 100 random bases receives a D on 2.6 and 11.2 % of reads, against 88–93 % without either mechanism. Plausibility constraints alone, without the cost, remove the chance Ds but also every real D at an open end. Chance Ds within about 30 bases of the V

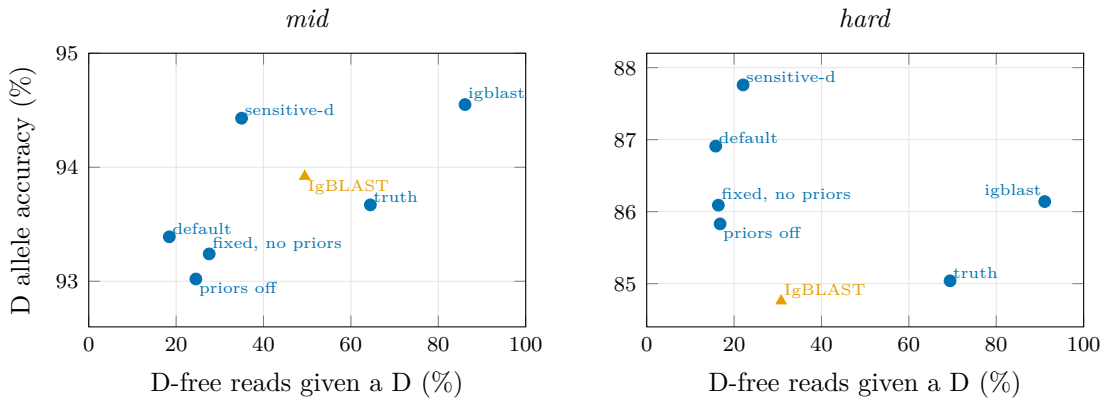


Figure 4: D accuracy against D-free specificity on IgSim reads. Circles are swig settings and calling profiles, the triangle is IgBLAST 1.22. A tool can buy D accuracy by calling a D more often; better models lie up and to the left.

remain (68 and 77 %): they sit where a real D of a truncated read sits, and no length or window term can separate them.

8.5 What each modelling term contributes

Table 6 switches each term off on the IgSim *mid* and *hard* sets and compares with the default read by read. The D trim and N/P priors raise D accuracy on both presets (+0.37 and +1.08 points; $p = 0.02$ and $3 \cdot 10^{-9}$) and, on *mid*, also leave 6.1 points more D-free reads without a D. The fixed $(2, -3)$ scale (`--calls fixed`) cannot carry the D trim and N/P terms, so its row changes the scale and drops those priors together. Against the run with the priors off, which differs only in the call model, calling at each read's own divergence leaves 3.0 points more D-free reads without a D on *mid* (24.5 against 27.6 %; 0.4 fewer on *hard*) and changes D by −0.2 and −0.3 points (93.02 against 93.24 % on *mid*, 85.83 against 86.09 % on *hard*). Its value lies in making the call a mode of the reported posterior, in D-free specificity on *mid*, and in the D priors it makes possible, not in D accuracy by itself. The open-flank cost and the conserved-anchor D window change essentially nothing on full-length reads: they act on reads that stop in or start after the junction, which the *mid* and *hard* presets rarely produce (Table 5 for the cost). The max-marginal D decoder changes the D allele of a few dozen full-length reads but gains exactly as many as it loses on IgSim (6 against 6 on *mid*, 26 against 26 on *hard*); on GenAIRR reads it adds up to 0.2 points of IGH D (95.55 against 95.42 at 5 % SHM), too little to make it the default. Skipping posteriors leaves every call unchanged. Raising the prior odds of a D (`sensitive-d`, prior odds 2) gains 0.8–1.0 points of D for 6–17 points more D-free reads given a D (Figure 4). The profiles without a D prior give a D to 64–91 % of D-free reads. On *hard* both are dominated and lose 0.8–1.9 points of D; on *mid* `sensitive-d` dominates the `truth` profile, and the `igblast` profile's 0.1-point edge in D over `sensitive-d` (94.55 against 94.43 %) costs 51 points more D-free reads given a D.

8.6 Where the remaining errors are

Table 7 attributes every wrong or missing D call on 10,000 IgSim reads per preset to the step where it arose. With a remnant of at least 8 nucleotides, the true D is almost never outside the search windows (0.04 % on *hard*) and almost never covered by the called J or V, the failure mode of sequential annotation. Most errors are adjudication errors: a placement of a correct allele existed, but the path chose another allele. These are short, mutated remnants that another allele explains as well or better under the model. On remnants of 1–7 nucleotides about half of the errors (53 and 58 %) are a missing D: in 38 and 40 % of the errors adjudication declined a

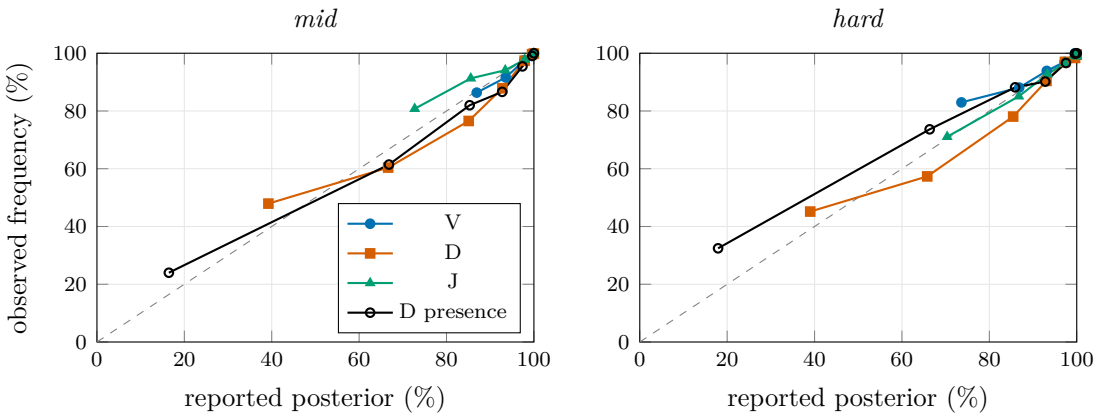


Figure 5: Reliability of the reported posteriors on IgSim reads: mean reported probability against observed frequency of the event, per bin (bins with at least 20 reads; most reads fall in the top bins). The event of a segment is "the reported names include the read's allele up to germline equivalence", that of D presence "the read has a simulated D".

D although a placement of an equivalent allele existed, as the D prior intends for evidence this weak, and in 16 and 17 % no such placement survived the D search.

The same holds for V and J. Aligning every allele unbanded under the same objective (an exhaustive oracle) found no V search error in 2,000 reads per preset (from the same sample as Table 7) and 1 and 3 J search errors on *mid* and *hard*; the remaining V errors are ties (26 and 84 reads) and model errors (12 and 68), in which the exhaustive optimum itself is not the truth because mutations fall on the positions that separate alleles.

8.7 Calibration

Posteriors are well calibrated wherever the candidates contain the truth (Figure 5). The expected calibration error is at most 0.5 % for V and 0.8 % for J on the *easy*, *mid* and *hard* presets, and 1.9 % for the D allele and 1.1 % for D presence on *hard* (Table 8). The D-allele posterior is over-confident in the 0.5–0.95 range: reported probabilities of 85 % correspond to 77–78 % observed. At 30 % SHM the V and J posteriors are over-confident (ECE 9.8 and 13.2 %): when retrieval misses the true gene, no candidate can carry its mass.

8.8 Constant region

On 10,000 IgSim reads per preset (both strands), about 80 % of which continue into one of the 101 human IGHC alleles with a complete coding start, the primary C call is anchored-equivalent to the truth on 98.5–99.2 % of reads with 16–31 nucleotides of C and on at least 99.65 % from 32 nucleotides, and C was called on at most 0.05 % of the about 2,000 reads per preset that stop before C. C parts of 8–15 nucleotides are called on 53–60 % of reads, as the 12-column floor intends. V, D and J calls are identical with and without the C stage by construction.

8.9 Speed

Table 9 gives throughput on the 10,000-read IgSim sets, as wall clock of the whole command, including database preparation (about 40 ms for the IMGT IGH set, every allele annotated on load). On four threads swig assigns 6,800–7,600 reads per second with posteriors, 76–88 times IgBLAST's throughput on the same reads and machine, and 8,300–10,200 without them; posteriors add 22–35 % to the run time. On the four out-of-sample GenAIRR IGH sets (Table 3; machine lightly loaded) IgBLAST took 81–90 s per 10,000 reads and swig 1.0–1.4 s. The SIMD kernel gives 2.3–2.7 times the throughput of the scalar kernel on one thread (2.3–2.5 for the

default AVX2 kernel across its two timings); AVX-512BW's 32 lanes ran 2–7 % faster than AVX2's 16, a difference of the order of the run-to-run variation (about 5 %). Every backend and every thread count produced byte-identical output on both sets.

9 Discussion

swig replaces the sequence of local decisions of a conventional annotator with one objective and its exact maximiser. Three consequences stand out in the results. First, the failure mode that motivates joint placement, a neighbouring segment absorbing a short D, has essentially disappeared (Table 7); the D errors that remain are choices between similar alleles on short, mutated remnants, where the evidence is present but weak. That is a modelling question, and the D trim and N/P priors show that modelling, not search, is where further accuracy lies: moving two cut-local prior terms into the graph raised IGH D accuracy on GenAIRR reads by up to 3.1 points at no measurable cost in speed. Second, because every non-alignment term is a log prior odds with a stated derivation, the trade-off between finding short D remnants and leaving D-free junctions alone has a single, interpretable knob, the prior odds of a D (Figure 4), rather than a set of thresholds. Third, the posterior is a by-product of the same recursion and the call is its mode, so the reported probabilities describe the call that is reported; they are well calibrated up to IgG-like mutation levels.

The pair of simulators matters. GenAIRR scores calls against tie sets that count only the bases a read actually shows, which rewards reporting genuine ties and penalises over-reporting; IgSim scores germline equivalence and separates D accuracy from D-free specificity and remnant length. On both, the IgBLAST comparison favours swig on V and D up to IgG-like mutation, except D on IgM-like reads, where IgBLAST's lead comes from 1–3-nt remnants. On IgSim, IgBLAST reports more exact junctions because swig withholds a junction whose anchor codon its alignments do not reach, and at 30 % SHM IgBLAST's V calls are better.

9.1 Limitations

- *All results are on simulated reads.* Both simulators make simplifying assumptions: uniform allele usage, independent trims, no chimeras or long adapters, and in IgSim no hotspot-targeted SHM and no codon-length indels. Several design parameters were measured on simulated reads: the open-flank allowance from simulated junction lengths, the D-cost credit on IgSim, and the D trim and N/P priors from an analysis of the previous release's losses on the GenAIRR sets of Table 2. The priors' means (5, 7 and 8 bases) are generic rather than fitted, and GenAIRR's own trim tables did no better in the prototype; on GenAIRR reads disjoint from the design sets the priors gained as much (Table 3), but both come from one simulator. Validation on real repertoires, for example donors with sequenced germline genotypes, is the most important next step.

- *Comparisons.* We did not compare with partis, IMGT/V-QUEST or MiXCR 4 (which requires a licence key).

- *The plausibility constraints are inherited thresholds*, not terms of the likelihood, and the call under them is the best admissible subset optimum rather than an exact maximum over every admissible path. They cost 0.4–2.9 points of junction exactness on simulated reads (Table 6) in exchange for robustness on truncated, chimeric and cross-locus reads that the simulators do not produce. A model of truncation and template switching would price what they forbid.

- *D on D-free junctions.* 16–18 % of IgSim reads without a D remnant still receive one, and a chance D within about 30 bases of a truncated read's V is indistinguishable from a real

D there (Table 5). On TRB, whose two D genes are short and rarely trimmed as much as the prior expects, the D-cost credit gives D-free reads a D more often.

- *Extreme mutation.* At 30 % SHM the strand test and seed retrieval lose intact k-mers: 4.4 % of reads are left uncalled, most of them (3.5 % of all reads) because the wrong strand was chosen, V accuracy falls below IgBLAST's, and the V and J posteriors are over-confident because the true gene is no longer among the candidates.
- *Statistics.* E-values use exact ungapped λ and K (anti-conservative for gapped scores) and the total database length (conservative for near-identical alleles); they gate hits and are not calibrated p-values. One divergence per read assumes uniform mutation across V, D and J.
- *One alignment per D allele and window.* A D allele's optimal local alignment in its window need not be its best clip at the chosen cuts; this is the one remaining approximation of the segmentation model.

9.2 Future work

The D-presence residual of the posterior tracks junction length, which points to an N/P length prior that decides D presence and not only re-ranks Ds. Allele usage is uniform in the model; usage priors learned from the repertoire itself, iterated with the calls, would matter where one gene dominates. Gapped Karlin–Altschul parameters and an effective database size can be fitted by simulation. Above all, the model's assumptions need testing on real data.

10 Conclusion

swig shows that germline assignment can be posed as one likelihood ratio and solved exactly, fast enough for whole repertoires: an exact semi-Markov recursion over every pair of cuts between neighbouring segments, priors derived as prior odds over the hypotheses each segment competes with, calls valued at each read's own divergence and posteriors from the same recursion. Every search shortcut carries a stated guarantee, and the remaining errors are located by construction. On GenAIRR reads it is the most accurate of the tools compared on V, D and J on every immunoglobulin set and on D for TRB, and on human IGH reads disjoint from those its D priors were designed on it still leads IgBLAST on V, D and J; on IgSim reads it leads IgBLAST on V and D up to IgG-like mutation (except IgM-like D, where IgBLAST's lead comes from unidentifiable 1–3-nt remnants) and trails on junction exactness and on V at 30 % SHM. It runs nearly two orders of magnitude faster than IgBLAST.

Availability

swig 0.39.7 is free software under the MIT licence, available at `https://github.com/mashu/swig` (Rust crate in `engine/`). It builds with `cargo build --release` on Rust $\geq$ 1.89, has no dependencies, and runs on x86-64 and AArch64 and as WebAssembly. The command `swig --v V.fasta --d D.fasta --j J.fasta reads.fastq` writes AIRR Rearrangement TSV; `swig simulate` and `swig evaluate` reproduce the IgSim evaluation for any germline set and any tool's AIRR output. The scripts and settings that produce every swig and IgBLAST number in this paper are distributed with the source.

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To be added.

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Table 1: Search steps and what each can lose.

Step	Guarantee	Rule and justification
Strand	heuristic	Count read positions whose 11-mer occurs in the V/J germline on each strand; trusted when the winner exceeds its Poisson null $\lambda_0 + 3\sqrt{\lambda_0}$ (λ_0: the read's k-mer count times the germline's 11-mer density), otherwise compare V seed votes on both strands.
V retrieval	heuristic	Seed votes per gene frame (9-mers weight 4; 7-mers weight 1 while the leader holds fewer than 25 nine-mer seeds). Keep 3 genes, and up to 16 while their votes are within $z\sqrt{2 \cdot 4 \cdot L}$ of the leader's L, $z = 2$: about two standard deviations of the difference of two seed counts, so the margin is proportionally wider for weakly supported, mutated reads. Every allele of a kept gene is a candidate.
V band	bounded	The seed cluster's diagonal range ± 8. A frame's runner-up cluster is taken as the far side of an indel, and the band spans both, when the two span at most 64 diagonals, the runner-up holds at least the votes of 3 nine-mers and 1/8 of the frame's best cluster, and that cluster those of 25. Holds unless a longer indel is not joined this way: its far side is unseeded or below those shares (e.g. an indel 20 bases from a V end), the two span more than 64 diagonals, or the frame's best cluster is weakly seeded.
V, J scoring	lossless	One representative R per gene frame is aligned; a sibling A satisfies $\underline{L} \leq S^*(A) \leq S^*(R) + \Delta(R, A)$ and is skipped when the upper bound is below the third-largest lower bound. The ranking from the third score up equals full scoring.
V/J keep	defined	Top 3 by extended and top 3 by local score with $E \leq 1$, plus ties.
J	bounded	Every J allele of the anchoring V's locus, unbanded, from 12 bases before the earliest V end to at most 225 bases plus the longest J past the last anchoring V's end; the whole read when no V is an anchor.
D windows	bounded	[V end $-$ 12, J start + 12) from the best anchoring V and the best J placement, plus the conserved-anchor window [Cys104 + 3, Trp/Phe118); an open side reaches 60 bases.
D prefilter	lossless	An allele is aligned only if it shares an ℓ-mer with the window, ℓ derived from the admission gate (exact run ≥ 4).
D alignment	approximate	One optimal local alignment per allele and window, the best 32 admissible ones (plus ties) per window; its clip at the chosen cuts need not be the best clip of that allele.
Adjudication	exact	Theorem 1.
Guards	defined	Best admissible subset optimum (Section 4).
C	exact within 7 bases	Anchored DP at the J exit (Section 5.1).

Table 2: Accuracy against GenAIRR's ambiguity-aware truth (mean Jaccard score of the reported allele set against the truth's tie set, %), 10,000 reads per set. MiXCR counts every hit tied with its top score; its figures come from an earlier run of the benchmark on the same reads. Bold: best of the row. A dash: the locus has no D.

	IgBLAST 1.22			MiXCR 3.0.13			swig 0.39.7		
Set	V	D	J	V	D	J	V	D	J
IGH, no SHM	99.86	97.80	99.48	99.76	94.56	99.53	**99.98**	**98.47**	**99.76**
IGH, 2 % SHM	99.80	96.55	99.36	99.65	93.34	99.37	**99.85**	**97.28**	**99.70**
IGH, 5 % SHM	99.64	93.24	99.14	99.30	90.22	98.81	**99.67**	**95.42**	**99.47**
IGH, 10 % SHM	99.36	84.28	98.48	98.45	81.07	96.60	**99.43**	**90.24**	**99.06**
IGH, realistic	99.53	92.56	99.05	99.01	89.55	98.73	**99.60**	**94.96**	**99.44**
IGK, 5 % SHM	99.63	–	98.88	99.14	–	98.35	**99.74**	–	**99.43**
IGK, realistic	99.48	–	98.11	97.99	–	97.09	**99.63**	–	**99.02**
IGL, 5 % SHM	99.53	–	98.32	98.48	–	98.16	**99.63**	–	**99.10**
IGL, realistic	99.34	–	98.32	97.16	–	96.49	**99.51**	–	**98.49**
TRB, no SHM	**99.97**	98.17	99.93	99.94	94.10	**100.00**	**99.97**	**99.91**	99.89
TRB, realistic	**99.94**	97.65	**99.92**	99.80	93.66	**99.92**	**99.94**	**99.50**	99.87

Table 3: Out-of-sample check: GenAIRR human IGH reads disjoint from those of Table 2 (10,000 per scenario; mean Jaccard score, %). "Priors off": swig 0.39.7 with `--d-trim off --np-length off`.

	swig 0.39.7			priors off	IgBLAST 1.22		
Scenario	V	D	J	D	V	D	J
no SHM	99.98	98.40	99.65	97.89	99.82	97.54	99.48
5 % SHM	99.66	94.97	99.39	92.71	99.59	92.65	99.15
10 % SHM	99.46	90.20	98.97	86.82	99.44	83.94	98.47
realistic	99.54	94.75	99.33	92.43	99.44	92.12	99.13

Table 4: IgSim reads against the IMGT human IGH set (germline-equivalent accuracy, %), forward strand; 10,000 reads per preset and 3,000 at 30 % SHM. "D, $\geq$ 8": reads whose D remnant is at least 8 nt. "D-free given a D": share of reads without a D remnant that receive a D call (lower is better). Bold: significantly better in a paired comparison (exact McNemar $p < 0.01$); V-gene accuracy was tested separately with the same rule.

Preset	Tool	V allele	V gene	D allele	D, ≥ 8	D-free given a D	J allele	Junction
easy	IgBLAST	99.47	99.97	99.26	99.37	0.0	99.93	100.00
	swig	**99.94**	100.00	**99.87**	**99.88**	0.0	99.93	100.00
mid	IgBLAST	98.06	99.90	93.92	97.28	49.4	98.75	**99.94**
	swig	**98.57**	99.96	93.39	**98.01**	**18.4**	98.75	99.53
hard	IgBLAST	92.02	99.56	84.76	88.24	30.8	92.06	**98.33**
	swig	**92.63**	99.62	**86.91**	**91.70**	**15.8**	**92.67**	96.81
30 % SHM	IgBLAST	**64.97**	**95.03**	35.74	35.82	80.6	40.60	**52.46**
	swig	61.60	89.67	**43.03**	**44.57**	**53.8**	**54.20**	46.01

Table 5: Truncated IgSim reads (2,000 per set and preset): right D calls on reads that hold a D, and any D call on D-free constructs. "Cost" is the open-flank cost, "guards" the plausibility constraints; the default has both.

Set	Read	Measure	default	cost off	guards off	both off
mid						
dcut	V N D, cut at the D's end	right D (%)	88.0	0.0	86.5	86.5
cdr3start	N D N J…, V removed	right D (%)	93.6	0.0	93.5	93.5
vcut30	V + 30 random bases	any D (%)	67.5	0.1	66.0	66.0
vcut100	V + 100 random bases	any D (%)	2.6	0.2	5.7	88.3
jcut100	100 random bases + J…	any D (%)	9.7	0.1	10.4	91.8
hard						
dcut	V N D, cut at the D's end	right D (%)	81.5	0.0	79.3	79.3
cdr3start	N D N J…, V removed	right D (%)	83.2	0.0	83.2	83.2
vcut30	V + 30 random bases	any D (%)	76.6	0.2	74.4	74.5
vcut100	V + 100 random bases	any D (%)	11.2	0.8	14.8	93.2
jcut100	100 random bases + J…	any D (%)	27.2	0.0	28.2	90.6

Table 6: Ablations on IgSim (10,000 reads per preset). Δ: change against the default in percentage points (positive: better), paired on the same reads; "0": no read changes correctness; "n.s.": $p \geq 0.05$ (exact McNemar). "D-free given a D" and the D-presence Brier score are absolute (default: 18.4 % and 0.0217 on *mid*, 15.8 % and 0.0227 on *hard*).

	mid				*hard*			
Change	ΔD	D-free given a D	Δjunction	Brier D pres.	ΔD	D-free given a D	Δjunction	Brier D pres.
D trim and N/P priors off	−0.37	24.5	n.s.	0.0240	−1.08	16.8	n.s.	0.0218
fixed-scale call (no D priors)	n.s.	27.6	+0.25	0.0224	−0.82	16.4	+0.28	0.0222
open-flank cost off	0	18.4	0	0.0217	−0.06	15.8	0	0.0239
plausibility constraints off	0	18.4	+0.44	0.0217	n.s.	15.8	+2.85	0.0223
conserved-anchor window off	n.s.	18.4	n.s.	0.0216	n.s.	16.0	0	0.0222
max-marginal D allele	n.s.	18.4	n.s.	0.0217	n.s.	15.8	0	0.0227
no posteriors	0	18.4	0	–	0	15.8	0	–
profile `sensitive-d`	+1.03	35.0	0	0.0232	+0.84	22.1	n.s.	0.0217
profile `truth`	n.s.	64.5	−0.39	0.0423	−1.87	69.4	−1.52	0.0417
profile `igblast`	+1.16	86.1	−0.30	0.0474	−0.78	91.1	−2.21	0.0458

Table 7: Attribution of D errors (% of reads with a D remnant in the class), on a separate sample of 10,000 IgSim reads per preset (Tables 4, 6 and 8 use another sample). "Adjudication": a placement of an equivalent allele existed but the path took another D or none; "not found": no such placement survived the D search.

	remnant ≥ 8 nt		remnant 1–7 nt	
	mid	*hard*	*mid*	*hard*
reads	8,535	8,483	974	1,006
D correct	97.90	91.76	51.85	44.83
wrong D, adjudication	1.91	6.90	17.25	16.60
wrong D, not found	0.12	0.83	5.34	6.76
wrong D, outside the windows	0.00	0.04	0.00	0.00
missing D, adjudication	0.03	0.23	18.07	22.27
missing D, not found	0.04	0.26	7.50	9.54
called J covers ≥ 3 bases of the true D	0.04	0.14	0.51	0.99

Table 8: Calibration of the reported posteriors on IgSim reads: expected calibration error (%) and Brier score.

	ECE (%)				Brier			
Preset	V	D	J	D presence	V	D	J	D presence
easy	0.09	0.20	0.05	0.04	0.0004	0.0010	0.0006	0.0001
mid	0.12	0.96	0.08	0.76	0.0064	0.0292	0.0063	0.0217
hard	0.47	1.87	0.80	1.14	0.0278	0.0556	0.0343	0.0227
30 % SHM	9.78	20.91	13.22	2.53	0.1654	0.2340	0.1978	0.0488

Table 9: Throughput on 10,000 IgSim reads (reads per second; wall clock of the whole command including database preparation; median of three runs for swig). The default kernel on this CPU is AVX2: the default and AVX2 one-thread rows are the same configuration timed separately and differ by run-to-run variation (about 5 %).

Configuration	*mid*	*hard*
swig, 4 threads	7,552	6,774
swig, 4 threads, no posteriors	10,211	8,272
swig, 1 thread	1,986	2,053
swig, 1 thread, no posteriors	2,690	2,515
swig, 1 thread, AVX-512BW (32 lanes)	2,235	2,188
swig, 1 thread, AVX2 (16 lanes)	2,089	2,152
swig, 1 thread, portable lanes (16)	1,575	1,612
swig, 1 thread, scalar kernel	842	877
IgBLAST 1.22, 4 threads	85.5	89.6